ExploratoryGenius

```
# Load in jsonlite package
library(jsonlite)
library(dplyr)
```

```
Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

    filter, lag

The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

```
library(igraph)
```

```
Attaching package: 'igraph'

The following objects are masked from 'package:dplyr':

    as_data_frame, groups, union

The following objects are masked from 'package:stats':

    decompose, spectrum
```

```
The following object is masked from 'package:base':

    union
```

```
#Assign Node Data and get a snapshot

#datausers_nominal <- stream_in(file("/Users/alexgould/Desktop/STA585GeniusProject/genius-exp
#head(datausers)
#data_annotate <- stream_in(file(("//Users/alexgould/Desktop/STA585GeniusProject/genius-exper
#head(data_annotate)
#songinfo <-  stream_in(file(("/Users/alexgould/Desktop/STA585GeniusProject/genius-expertise/
#head(songinfo)
# save(datausers_nominal, data_annotate, songinfo, file = "/Users/alexgould/Desktop/STA585Ge

# Load in Node data
load("/Users/alexgould/Desktop/STA585GeniusProject/initialfiles.Rdata")

# Assign Edge Data
edges_nominal <- read.csv("/Users/alexgould/Desktop/STA585GeniusProject/genius-expertise/foll
length(unique(edges_nominal$follower))
```

```
[1] 118199
```

```
length(unique(edges_nominal$following))
```

```
[1] 737206
```

```
#Make Dataset with only users in node dataset and who are also contained
# in folower and following

valid_users <- intersect(
  unique(data_annotate$user[!is.na(data_annotate$user)]),
  intersect(
    intersect(unique(datausers_nominal$url_name),
              unique(edges_nominal$follower)),
    unique(edges_nominal$following)
  )
)
edges_reduced <- data.frame()
edges_reduced <- edges_nominal[edges_nominal$follower %in% valid_users &
                                 edges_nominal$following %in% valid_users,
```

```
                                c("follower", "following")]
#stopifnot(all(unlist(edges_reduced) %in% nodes_reduced$url_name))
dim(edges_reduced)
```

```
[1] 15884     2
```

```
#Make sure we intersect the annotation dataset and nodes dataset because annotate dataset has
valid_nodes <- datausers_nominal[datausers_nominal$url_name %in% valid_users, ]

valid_annotate <- data_annotate[data_annotate$user %in% valid_users, ]

# Merge nodes and annotate by user name
nodes_reduced <- merge(valid_nodes, valid_annotate, by.x = "url_name",
                       by.y = "user", all.x = TRUE)
dim(nodes_reduced)
```

```
[1] 24133    18
```

```
#Delete duplicate rows
nodes_reduced <-nodes_reduced[!duplicated(nodes_reduced$url_name),]

#Check class of each variable in the merged dataset
sapply(nodes_reduced, class)
```

```
         url_name                iq             stats             roles
      "character"         "integer"       "data.frame"            "list"
    followers_num      following_num              time              type
        "integer"         "integer"       "character"       "character"
            votes            pyongs          acceptor               url
        "integer"         "integer"       "character"       "character"
contribution_stats         edits_lst              song            lyrics
           "list"            "list"       "character"       "character"
           artist           content
      "character"       "character"
```

```
## Split up with dataframe Stats variable
nodes_reduced$annotations <- nodes_reduced$stats$Annotations
nodes_reduced$suggestions <- nodes_reduced$stats$Suggestions
nodes_reduced$transcriptions <- nodes_reduced$stats$Transcriptions
nodes_reduced$questions <- nodes_reduced$stats$Questions
```

```r
nodes_reduced$answers <- nodes_reduced$stats$Answers
nodes_reduced$pyongs <- nodes_reduced$stats$Pyongs

nodes_reduced <- subset(nodes_reduced, select = -stats)

## Deal with list variables
class(nodes_reduced$roles)
```

```
[1] "list"
```

```r
class(nodes_reduced$contribution_stats)
```

```
[1] "list"
```

```r
class(nodes_reduced$edits_lst)
```

```
[1] "list"
```

```r
#Split all roles into Boolean variables
all_roles <- sort(unique(unlist(nodes_reduced$roles)))
for (r in all_roles) {
  col <- paste0("role_", make.names(r))
  nodes_reduced[[col]] <- vapply(nodes_reduced$roles, function(x) r %in% x, logical(1))
}
#Delete original roles
nodes_reduced <- subset(nodes_reduced, select = -roles)

#Check contribution_stats
head(nodes_reduced$contribution_stats)
```

```
[[1]]
NULL

[[2]]
NULL

[[3]]
NULL
```

```
[[4]]
NULL

[[5]]
NULL

[[6]]
NULL
```

```
length(!is.null((nodes_reduced$edits_list)))
```

```
[1] 1
```

```
#Most contribution stats values are null in this subset, delete from dataframe

nodes_reduced <- subset(nodes_reduced, select = -contribution_stats)

#Check edits_lst
head(nodes_reduced$edits_lst)
```

```
[[1]]
NULL

[[2]]
NULL

[[3]]
NULL

[[4]]
NULL

[[5]]
NULL

[[6]]
NULL
```

```
unlist(nodes_reduced$edits_list)
```

```
NULL
```

```r
#All edits_lst values are null in this subset, delete from dataframe
nodes_reduced <- subset(nodes_reduced, select = -edits_lst)

#Subset the songinfo dataset by unlisting the tags (genre) and set the most common genre as t
top_tag <- function(x) {
  x <- unlist(x)
  if (length(x) == 0) NA_character_ else x[1]
}
song_tags <- data.frame(
  key  = tolower(songinfo$url_name),
  tag  = vapply(songinfo$tags, top_tag, character(1)),
  stringsAsFactors = FALSE
)
song_tags <- song_tags[!duplicated(song_tags$key), ]   # one row per song

#Merge song_tags to nodes_reduced
nodes_reduced$tag <-song_tags$tag[match(tolower(nodes_reduced$song), song_tags$key)]

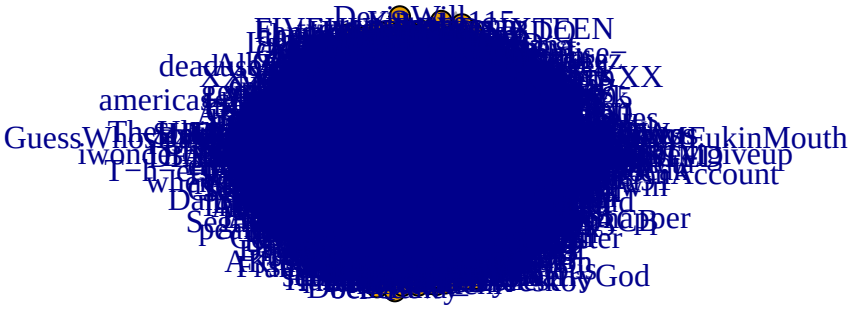

#New colnames
colnames(nodes_reduced)
```

```
 [1] "url_name"         "iq"               "followers_num"    "following_num"
 [5] "time"             "type"             "votes"            "pyongs"
 [9] "acceptor"         "url"              "song"             "lyrics"
[13] "artist"           "content"          "annotations"      "suggestions"
[17] "transcriptions"   "questions"        "answers"          "role_Contributor"
[21] "role_Editor"      "role_Mediator"    "role_Moderator"   "role_Staff"
[25] "tag"
```

```r
dim(nodes_reduced)
```

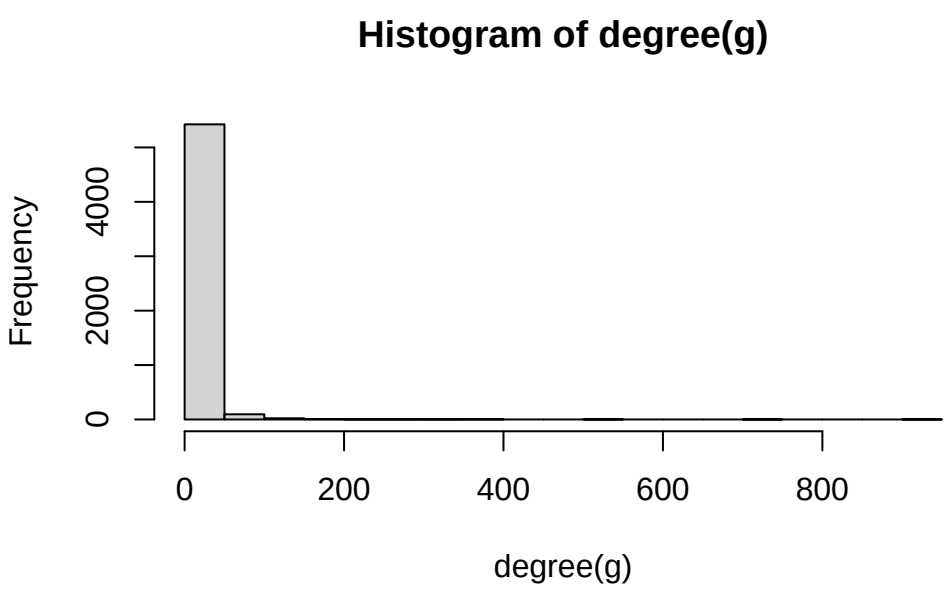
```
[1] 5559   25
```

```r
#Create preliminary graph
g <- graph_from_data_frame(vertices = nodes_reduced, d= edges_reduced, directed = TRUE)
plot(g)
```



Descriptive Analysis

```
# A. Node Degree Distribution
hist(degree(g))
```




```
mean(degree(g))
```

```
[1] 5.714697
```

```
sd(degree(g))
```

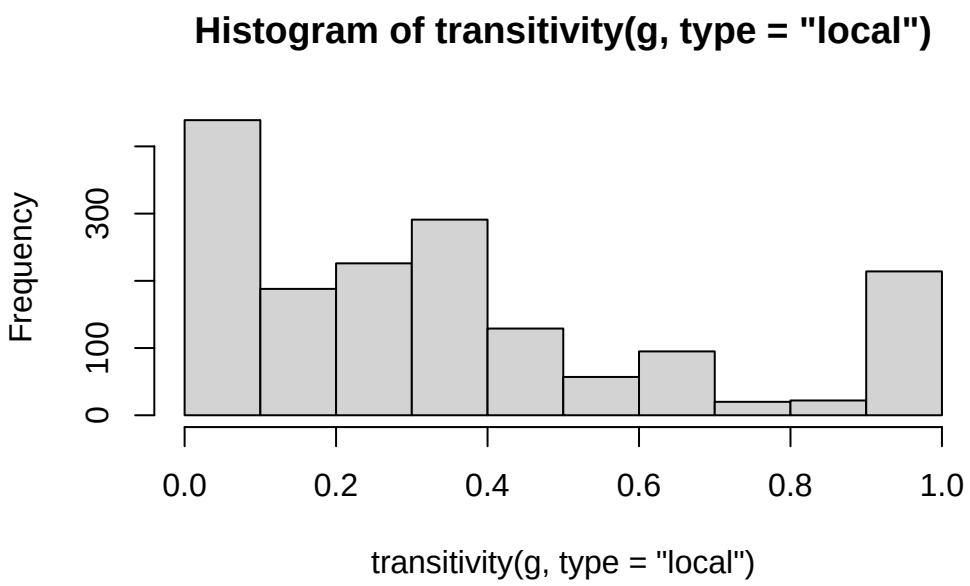
```
[1] 25.49265
```

```r
# B. Global and Local Density
#Global
edge_density(g)
```

```
[1] 0.0005140965
```

```r
#Local


#C. Clustering Coefficient
hist(transitivity(g, type = "local"))
```




```r
transitivity(g, type = "global")
```

```
[1] 0.09051928
```

```r
mean(na.omit(transitivity(g, type = "local")))
```

```
[1] 0.3544141
```

```r
sd(na.omit(transitivity(g, type = "local")))
```

```
[1] 0.3199898
```

```
# D: Average Betweenness
mean(betweenness(g))
```

```
[1] 1553.095
```

```
#E : Degree/Nominal Assortativity
#Degree Assortativity
assortativity_degree(g)
```

```
[1] -0.1909851
```

```
#Nominal Assortativity


#F: Graph Partinitioning
#cluster_fast_greedy(g)
#Only works on undirected graphs
```